Wei-Cheng Wang, Ph.D.

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Summary

Ph.D. in Computer Science Engineering with expertise in multimodal learning (audio, video), generative models, and privacy-preserving ML. Experienced in taking ML systems from clean lab benchmarks to unconstrained real-world environments, focusing on anomaly detection, domain shift, and privacy constraints in surveillance-like settings. Proven track record of resolving domain shift and data scarcity challenges through unsupervised techniques across both vision and acoustic sensing systems. Also experienced in building agentic pipelines that orchestrate large models and tools for surveillance-like applications and synthetic data generation. I am seeking Applied Scientist or Research Engineer roles to build robust, scalable, and trustworthy AI systems.

Education

Ph.D. *Ghent University, Belgium*
- *Computer Science Engineering* 2019-2025

Master of Science *National Cheng Kung University (NCKU), Taiwan*
- *Computer and Communication Engineering.* 2014-2016

Bachelor of Science *National Cheng Kung University (NCKU), Taiwan*
- *Electrical Engineering.* 2010-2014

Experience

Doctoral Researcher (*Ghent University, 2019-2025*)
- Conducted research on real-world urban surveillance and audio-visual monitoring systems, focusing on domain shift, data scarcity, and privacy leakage to enable robust deployment in unconstrained environments, resulting in 4 peer-reviewed journal publications.
- Collaborated with industry partner Asasense to analyze how state-of-the-art anomaly detection and sound tagging models behave on real-world audio, informing robustness requirements for deployment.
- Co-supervised a master's thesis on visual anonymization for traffic surveillance, advising on how to remove privacy-sensitive content (e.g., license plates) while keeping footage usable for downstream analytics.

Teaching Assistant (*Ghent University, 2021-2023*)
- Mentored graduate students on end-to-end ML projects, including image-text anomaly and fraud-like risk detection scenarios and gesture recognition built in official collaboration with Sony Depthsensing Solutions.

Co-President (*UGent Taiwanese Student Association, 2020-2022*)
- Led student organizations and cross-cultural initiatives during the pandemic, strengthening skills in coordination, communication, and crisis-time support.

Projects

Applied AI Systems: Job-Hunter Multi-Agent Co-pilot (*Self-initiated*, *2025-present*)
- Built a multi-agent AI system to reduce labor-intensive workload in job matching and to validate the orchestration logic for a future physically-aware synthetic surveillance dataset generator.
- Designed an orchestrated pipeline that leverages LLM APIs and text embeddings over heterogeneous inputs to analyze job postings, estimate match and effort, and retrieve reusable skills and experience patterns from previous resumes.
- Served as a sandbox to explore how to orchestrate large models and tools in practice, surfacing lessons on failure modes, latency, and prompt/embedding design that I now use to design agentic pipelines for LoRA-augmented vision models and synthetic anomaly dataset generation in surveillance-like settings.

Multi-modal Surveillance: From Lab to Street (*Ph.D. Research*, *2021-2025*)

- Developed scalable and privacy-friendly methods for deploying surveillance ML systems in unconstrained real-world environments, focusing on closing the gap between clean lab benchmarks and noisy, shifting operational data.
- Studied how state-of-the-art methods break down on real-world surveillance data and proposed practical solutions across unsupervised audio-visual representation learning, opt-in privacy-preserving frameworks, and source-free transferability assessment to specialize models efficiently for unknown target domains.
- Identified key real-world constraints: including false negatives and information bottlenecks in contrastive learning, unknown privacy attackers in the wild, and inaccessible source data with unlabeled target domains. Distilled practical guidelines for adapting and hardening multimodal models when moving from clean datasets to real operational smart city environments.

SensCity: Real-World Acoustic Surveillance (*Ph.D. Research*, *2019-2021*)

- Evaluated the robustness of state-of-the-art anomaly detection and sound tagging models when deployed on large-scale, unconstrained urban audio collected through the imec SenseCity program in collaboration with Asasense.
- Analyzed how performance degrades under real-world noise, overlapping events, and distribution shifts across locations and devices, and investigated model and pipeline choices that are more resilient in operational settings.
- Provided empirical insights and recommendations on bridging the gap between benchmark results and field performance for acoustic surveillance, informing how to stress-test and select models before city-scale deployment.

Early Applied Computer Vision and Prototype Transfer (*Pre-PhD R&D*, *2014-2018*)

- Conducted early research and engineering in computer vision during my M.Sc. and research assistantship, focusing on driver monitoring and surveillance video analysis, resulting in 6 publications including *IEEE TITS* and *IEEE TCSVT*.
- Led award-winning AI prototypes in international competitions (APICTA, InnoServe) with a student-only team, designing and presenting solutions competing against corporate teams.
- Translated research ideas into deployed prototypes with real stakeholders, including a multi-modal (ECG and facial features) driver monitoring system co-developed with national-level partner ARTC and a surveillance analytics prototype that was later transferred to an industry partner, building early experience in taking computer vision models beyond the lab into real-world usage.

Selected Publications

Embedding-based pair generation for contrastive representation learning in audio-visual surveillance data
Frontiers in Robotics and AI, 2025. (First author)

Source-free model transferability assessment for smart surveillance via randomly initialized networks
Sensors, 2025. (First author)

Privacy-preserving visual analysis: Training video obfuscation models without sensitive labels
Applied Intelligence, 2024. (Second author)

An opt-in framework for privacy protection in audio-based applications
IEEE Pervasive Computing, 2022. (First author)

Clustering trajectories in heterogeneous representations for video event detection
IEEE International Conference on Image Processing (ICIP), 2018. (First author)

For a complete list of publications, see: Google Scholar

Skills

Computer Vision, Multimodal Learning, Self-Supervised / Contrastive Learning, Generative Models, Privacy-Preserving ML, Data-Centric AI, Model Evaluation & Benchmarking, Domain Shift & Source-Free Transferability.	Expertise
Python, PyTorch, TensorFlow, Weights & Biases, TensorBoard, C++, OpenCV, Docker, Git.	Toolkit